

OMID ZANGANEH

Senior GIS Developer and Analyst

MS Geography — Geographic Information Science & Technology · Workflow Automation · AI/ML Integration

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PROFESSIONAL SUMMARY

Senior GIS Developer and Analyst with a track record of turning manual spatial workflows into reliable production tools. Specializes in ArcGIS Pro SDK (.NET/C#/WPF), Python geoprocessing, SQL Server pipelines, and applied AI for engineering teams. Proven results: ~90% fewer manual GIS steps, multi-day workflows reduced to minutes, and AI-assisted contract sourcing in hours instead of months. 2025 Edison Award Nominee and 2026 Edison Award Winner at Olsson. MS Geography (GIS&T), University of Nebraska at Omaha, 4.0 GPA.

WORK EXPERIENCE

Olsson ♦ 2025 Nominee & 2026 Edison Award Winner

March 2025 – Present

GIS Developer and Analyst

Lincoln, Nebraska

- Worked directly with many telecom and infrastructure clients, delivering GIS tools, network design support, and spatial analysis that improved speed and consistency of delivery.
- Built production Python and C# GIS apps, including bore profile automation that cut processing from days to minutes.
- Created an ArcGIS Automation Suite of Python toolboxes that reduced manual GIS steps by ~90% and sped up fiber network design.
- Shipped ArcGIS Pro add-ins (.NET/C#/WPF) adopted across projects and teams — including a GIS Data Downloader used firm-wide, FTTH Network Designer, RF Analysis panel, and a Street View tool for in-map pole and asset verification.
- Learned iBwave to deliver DAS (distributed antenna system) designs, and built AI tools (Azure OpenAI, Google AI Studio, YOLO) for contract sourcing and parcel-owner classification that spot development activity for fiber expansion analysis.

University of Nebraska at Omaha

January 2024 – August 2025

Graduate Teaching Assistant — Instructor of Record

Omaha, Nebraska

- Taught Human-Environment Geography labs as sole instructor of record to 150+ students across three semesters.

University of Nebraska at Omaha

June 2024 – August 2025

GIS Technician — Omaha Spatial Justice Project

Omaha, Nebraska

- Digitized historical parcels and legal records into a geodatabase of racially restrictive covenants in Douglas County.

SELECTED PROJECTS

RFP Radar — AI-Powered RFP Sourcing

Azure OpenAI · Google Gemini · Playwright · Python/C#

Desktop app for web-grounded RFP search, classification, and CSV export. Months of sourcing → hours.

Bore Profile Automation — Directional Drilling Profile Generator

Python · C# · ArcGIS Pro · SQL Server · Matplotlib

Reads spatial waypoints and elevation models; generates 2D/3D bore profiles. Days → minutes.

Fiber Automatic Expansion — Density Grouping & Fiber Build-Area Planning

C# · .NET · ArcGIS Pro SDK · WPF · Spatial Analysis

Grids the study area for density analysis, flood-fills viable build zones by PPM thresholds, and generates economic scorecards.

Used across multiple clients; days/weeks → minutes.

FTTH Network Designer — Automated Fiber Optic Network Planning

C# · .NET 8 · ArcGIS Pro SDK · WPF · Kruskal MST

Places shafts, connects homes, and builds an optimal Main Trunk / Terminal Branch backbone. Days → minutes.

ArcGIS Data Downloader Add-in — Multi-Source GIS Data Acquisition

C# · .NET 8 · ArcGIS Pro SDK · WPF · REST APIs

Downloads OSM, USGS, FEMA, Census/TIGER, Wikipedia, and BSL layers into ArcGIS Pro. Adopted firm-wide across projects and teams.

Street View in ArcGIS Pro — In-App Pole & Asset Verification

C# · .NET 8 · ArcGIS Pro SDK · WPF · WebView2

One-click Google Street View inside ArcGIS Pro so analysts can verify poles and street-level equipment without leaving the map.

Parcel Owner Classifier — AI Development Detection for Fiber Expansion

Python · Azure OpenAI · SQL Server · Tkinter · PyInstaller

Classifies parcel owners as development/investment firms in large SQL datasets to find where growth is happening for fiber expansion analysis.

RF Analysis Tool — 8-Tool Wireless Planning Panel

C# · ArcGIS Pro SDK · Python

Coverage prediction, PCI/RSI planner, tilt/azimuth optimizers, interference analysis, and tower placement — all as map layers.

Aerial & Streetview AI Detection — YOLO Utility Infrastructure Detection

Python · YOLO · OpenCV · Aerial/Street Imagery APIs

Detects and classifies utility assets from aerial and street imagery; exports georeferenced results to ArcGIS.

GeoPipe — Enterprise GIS ETL to SQL Server

Python · CustomTkinter · pyodbc · SQL Server · PyInstaller

GUI ETL for large spatial/tabular imports with schema auto-detection, GEOMETRY/GEOGRAPHY support, and connection-loss auto-resume.

omidzanganeh.com — Full-Stack Portfolio & Browser GIS Tools

Next.js 16 · React · TypeScript · Leaflet · Vercel

Personal site with 10+ browser GIS tools (geocoder, isochrone, elevation, census, coordinate converter), Gym Flow, Stock Screener, and an AI news aggregator.

EDUCATION

Master of Science, Geography — Geographic Information Science & TechnologyDecember 2025

University of Nebraska at Omaha, Nebraska **GPA: 4.00** · **GRACA Project Award**

Thesis: Spatiotemporal Analysis of NOx Emissions from U.S. Cement Plants Using TROPOMI Data — remote sensing, hotspot analysis, and environmental justice.

Bachelor of Science, Geomatics (Surveying) Engineering

August 2016

Geomatics College of National Cartographic Center (GCNCC), Tehran

TECHNICAL SKILLS

Languages & Frameworks Python · C# / .NET · SQL · TypeScript · JavaScript · HTML/CSS · React · Next.js · WPF

GIS & Spatial ArcGIS Pro · ArcGIS Pro SDK · ArcPy · Python Toolboxes (.pyt) · ArcGIS Online · ArcGIS Enterprise · QGIS · Geoprocessing · Network Analysis · Remote Sensing · Google Earth Engine · iBwave

AI / ML Azure OpenAI · Azure AI Foundry · Google AI Studio / Gemini · YOLO · OpenCV · Prompt Engineering · Web Grounding · Batch Classification Pipelines

Data & Backend SQL Server · Supabase · PostgreSQL · pyodbc · ETL Pipelines · Spatial Types (GEOMETRY/GEOGRAPHY) · Smartsheet API

Web & Cloud Next.js App Router · Vite · Leaflet · Vercel · REST APIs · PWA · Microsoft Azure · Google Cloud

Desktop & Tooling CustomTkinter · PyInstaller · Playwright · Matplotlib · AutoCAD · ENVI · SNAP · Photomod · Tableau · Adobe Photoshop · Adobe Illustrator

Languages English (Fluent) · Persian (Native)

For project write-ups, live GIS tools, and demos: omidzanganeh.com
